



Introduction to Machine Learning

NYU K12 STEM Education: Machine Learning

Department of Electrical and Computer Engineering,
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- ▶ [Course Website](#)
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Tell the class about... the person next to you

- ▶ Name
- ▶ (Rising) Grade Number
- ▶ City, Country
- ▶ Favorite Summer Activity/Hobby
- ▶ Why do you want to learn Machine Learning?

Outline

1. Course Outline

2. Introduction

3. Matrices

4. Vectors

5. Matrix Multiplication

6. Matrix Inverse

7. Python

Week 1

- ▶ Day 1 : Introduction to Machine Learning
- ▶ Day 2 : Supervised Learning - Linear Regression
- ▶ Day 3 : Over-fitting and Regularization
- ▶ Day 4 : Supervised Learning - Classification
- ▶ Day 5 : Mini Project

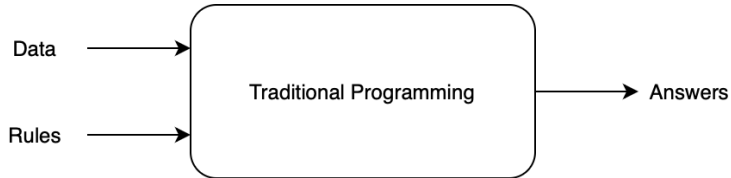
Week 2

- ▶ Day 6 : Introduction to Deep Learning and Neural Networks
- ▶ Day 7 : Convolutional Neural Networks
- ▶ Day 8 : Advanced Deep Learning Topics
- ▶ Day 9 : Ethics and Future of AI
- ▶ Day 10 : Final Project

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Traditional Programming



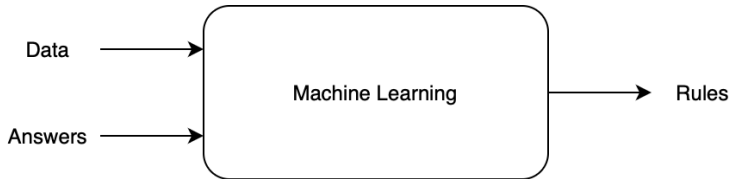
Traditional Programming

A large grid of small dots with the numbers 0, 8, 1, and 7 drawn in a thick, hand-drawn red line. The numbers are positioned in the center of the grid, with the 0 on the left, followed by the 8, then the 1, and finally the 7 on the right.

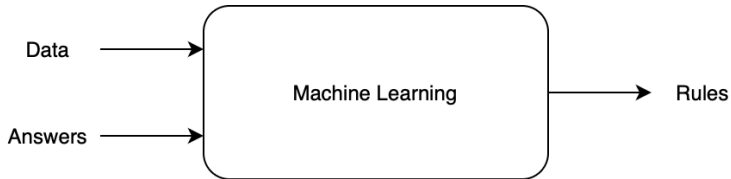
Challenges



Machine Learning



Machine Learning



Definition

Machine Learning is a field of study that gives computers the ability to learn without being explicitly programmed.

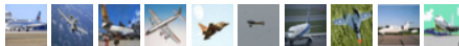
Example: Digit Recognition

A 10x15 grid of handwritten digits from 0 to 9. Each row contains 15 examples of a single digit. The digits are written in various styles, including standard, cursive, and stylized, to illustrate the variability in handwriting for digit recognition.

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

Example: Image Classification

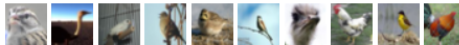
airplane



automobile



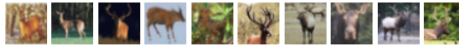
bird



cat



deer



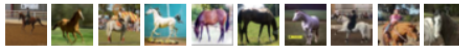
dog



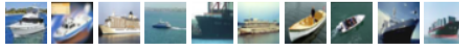
frog



horse



ship



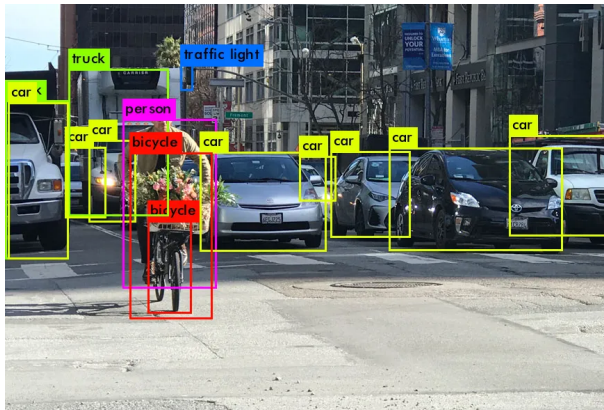
truck



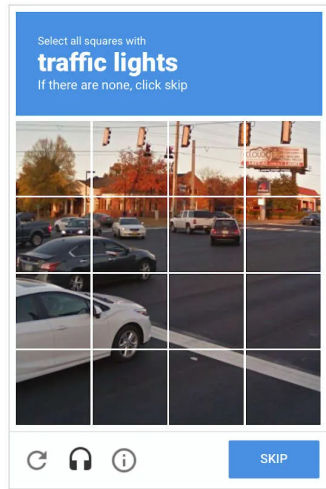
But why is Machine Learning important now?

- ▶ Big Data
 - Massive storage, large data centers
 - Massive connectivity
 - Sources of data from internet and elsewhere
- ▶ Computational advances
 - GPUs and hardware
 - Distributed machines, clusters

Labeled Data



How are labels generated?



Types of Machine Learning Problems

Supervised Learning

- ▶ Model is trained on labeled datasets
- ▶ Learn a mapping from inputs to outputs

Types of Machine Learning Problems

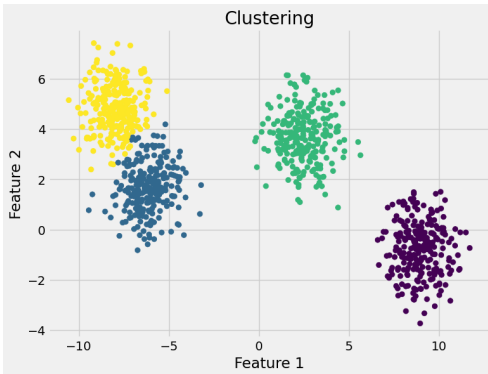
Supervised Learning

- ▶ Model is trained on labeled datasets
- ▶ Learn a mapping from inputs to outputs

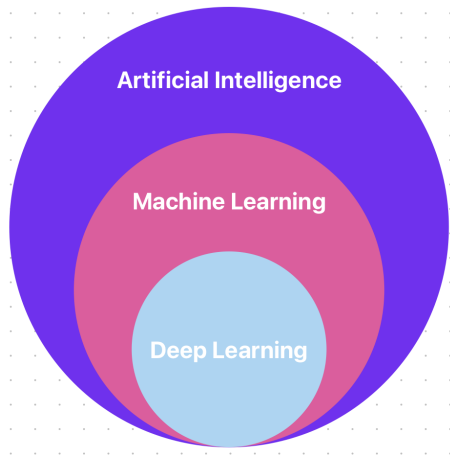
Unsupervised Learning

- ▶ Model is trained on unlabeled datasets
- ▶ Model infers patterns and structure from the data

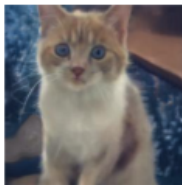
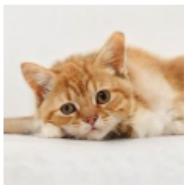
Unsupervised Learning Example



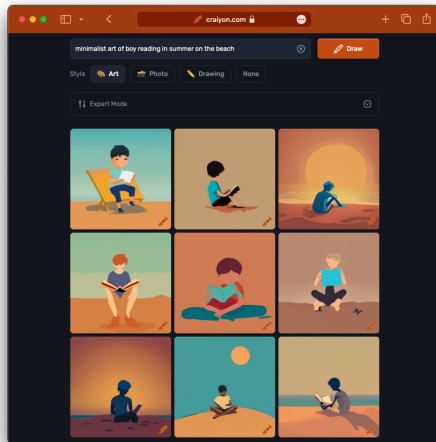
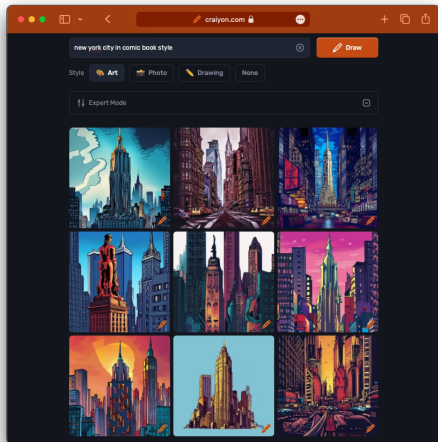
Artificial Intelligence vs. Machine Learning vs. Deep Learning



A break to look at cats



Example: Dall.E

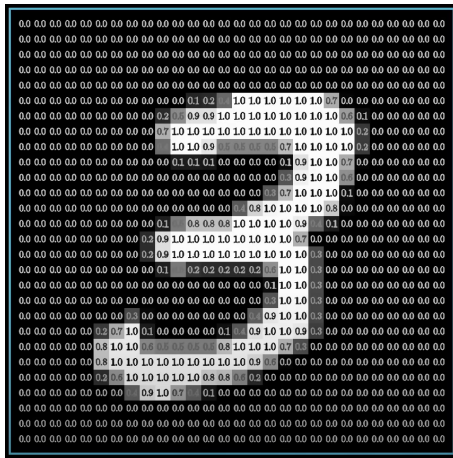


- ▶ Form Teams: Create groups of 4 students each.
- ▶ Assign Roles: Select one team member to be the guesser. The remaining three members will be the actors.
- ▶ Pick a Sentence: A random sentence will be drawn from the bag. The sentence should be hidden from the guesser.
- ▶ Act It Out: The actors will silently act out the sentence together.
- ▶ The guesser must try to guess the sentence based on the actors' performance.
- ▶ Each team has 2 minutes to make as many correct guesses as possible.

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4. Vectors
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6. Matrix Inverse
7. Python

Why do we learn Vectors and Matrices?



Matrices

A matrix is a rectangular array of numbers (or other mathematical objects).

Example:

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \quad M = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ -4 & 5 \end{bmatrix}$$

Size of a Matrix

The size of a matrix is defined by the number of rows and columns it contains.

Example:

Matrix A has 2 rows and 2 columns, hence the size of the matrix is 2×2

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$$

Size of a Matrix

Similarly, matrix M is of size ...?

$$M = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ -4 & 5 \end{bmatrix}$$

Size of a Matrix

In general, a matrix A of size $m \times n$ is given as,

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

where a_{ij} represents the i^{th} row and j^{th} column element.

Size of a Matrix

$$M = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ -4 & 5 \end{bmatrix}$$

► $m_{31} = \dots?$

► $m_{22} = \dots?$

Matrix Addition

Matrices of the same size may be added together, element-wise.

Example:

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 8 \\ 7 & 1 \end{bmatrix}$$

$$\therefore C = A + B = \begin{bmatrix} 1+0 & 1+8 \\ 2+7 & 1+1 \end{bmatrix} = \begin{bmatrix} 1 & 9 \\ 9 & 2 \end{bmatrix}$$

Matrix Subtraction

Similarly, matrices of the same size may be subtracted together, element-wise.

Example:

$$\therefore D = A - B = \begin{bmatrix} 1 - 0 & 1 - 8 \\ 2 - 7 & 1 - 1 \end{bmatrix} = \begin{bmatrix} 1 & -7 \\ -5 & 0 \end{bmatrix}$$

Matrix Scalar Multiplication

Matrices can be scaled by a number. The resulting matrix is computed element-wise. This operation is called scalar multiplication.

Example:

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \quad c = 3$$

$$\therefore c \cdot A = \begin{bmatrix} 1 \times 3 & 1 \times 3 \\ 2 \times 3 & 1 \times 3 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 6 & 3 \end{bmatrix}$$

Matrices Exercise

$$\begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 4 & 7 \end{bmatrix} + \begin{bmatrix} 9 & 2 \\ -7 & 6 \\ 3 & 1 \end{bmatrix} = \dots?$$

$$\begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 4 & 7 \end{bmatrix} - \begin{bmatrix} 9 & 2 \\ -7 & 6 \\ 3 & 1 \end{bmatrix} = \dots?$$

$$2 \cdot \begin{bmatrix} 1 & 9 \\ 3 & -2 \end{bmatrix} = \dots?$$

Transpose of a Matrix

The transpose of a matrix is formed by swapping rows and columns.

Example:

$$M = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ -4 & 5 \end{bmatrix}$$

$$\therefore M^T = \begin{bmatrix} 1 & 2 & -4 \\ 3 & -1 & 5 \end{bmatrix}$$

A transposed matrix is denoted as M^T

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Vectors

Matrices with a single row are called **row vectors**. A row vector is a $1 \times n$ matrix, consisting of a single row of n elements.

Example:

$$U = [1 \quad 2 \quad 3]$$

Vectors

Matrices with a single column are called **column vectors**. A column vector is a $n \times 1$ matrix, consisting of a single column of n elements.

(We consider column vectors by default)

Example:

$$V = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Vector Addition

Vectors of the same dimensions may be added together, element-wise.

Example:

$$V = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad W = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\therefore X = V + W = \begin{bmatrix} 1 + 4 \\ 2 + 5 \\ 3 + 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

Vector Subtraction

Similarly, vectors of the same dimensions may be subtracted together, element-wise.

Example:

$$\therefore Y = W - V = \begin{bmatrix} 4 - 1 \\ 5 - 2 \\ 6 - 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}$$

Vector Scalar Multiplication

Vectors can be scaled by a number. The resulting vector is computed element-wise.

Example:

$$V = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad c = 5$$

$$\therefore c \cdot V = \begin{bmatrix} 1 \times 5 \\ 2 \times 5 \\ 3 \times 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 15 \end{bmatrix}$$

Vector Dot Product

Vector Dot Product (a.k.a. **Inner Product**) is the sum of the products of the corresponding entries of the two vectors.

Example:

$$\therefore V \cdot W = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 1 \times 4 + 2 \times 5 + 3 \times 6 = 4 + 10 + 18 = 32$$

Norm of a Vector

The **norm of a vector** (l^2 -norm) for a vector $Z = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ is given as,

$$\|Z\|_2 = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

The l^2 -norm is denoted as $\|Z\|_2$ or $\|Z\|$

The **squared norm** is the square of the norm of a vector.

$$\|Z\|_2^2 = \left(\sqrt{3^2 + 4^2}\right)^2 = 25$$

The squared norm is denoted as $\|Z\|_2^2$ or $\|Z\|^2$

Vector Exercise

$$P = \begin{bmatrix} 3 \\ 2 \\ 9 \\ 4 \end{bmatrix} \quad Q = \begin{bmatrix} 1 \\ 9 \\ 0 \\ 3 \end{bmatrix}$$

- ▶ $3Q + 2P = \dots?$
- ▶ $Q \cdot Q = \dots?$
- ▶ $\|Q\|^2 = \dots?$
- ▶ $P \cdot Q = \dots?$
- ▶ $\|P\| \cdot \|Q\| = \dots?$

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Matrix Multiplication

Two matrices, A and B , can be multiplied together provided their shapes meet the criteria:

- ▶ Number of columns of A = Number of rows of B
- ▶ Result is a matrix of shape (Number of rows of A $\times$ Number of columns of B)

Matrix Multiplication

Example:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -3 & 1 \\ 0 & -1 \\ 3 & 1 \end{bmatrix}$$

Size of A is 2×3 , size of B is 3×2 . Therefore, the resulting matrix C is of size 2×2

$$A \times B = C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

The entries of C are given by the dot product of the corresponding row of A and the corresponding column of B .

Matrix Multiplication

$$c_{11} = \begin{bmatrix} 2 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 0 \\ 3 \end{bmatrix} = 2 \times -3 + -1 \times 0 + 0 \times 3 = -6$$

$$c_{12} = \begin{bmatrix} 2 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 2 \times 1 + -1 \times -1 + 0 \times 1 = 3$$

$$c_{21} = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 0 \\ 3 \end{bmatrix} = -1 \times -3 + 0 \times 0 + 1 \times 3 = 6$$

$$c_{22} = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = -1 \times 1 + 0 \times -1 + 1 \times 1 = 0$$

Matrix Multiplication

$$\therefore C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} -6 & 3 \\ 6 & 0 \end{bmatrix}$$

To sum up: Multiplication of two matrices is defined if and only if the number of columns of the left matrix is the same as the number of rows of the right matrix. If A is an $m \times n$ matrix and B is an $n \times p$ matrix, then their matrix product $A \times B$ is the $m \times p$ matrix whose entries are given by dot product of the corresponding row of A and the corresponding column of B .

Matrix Multiplication Exercise

$$\begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \times \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \dots?$$

$$\begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \times \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} = \dots?$$

$$\begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} = \dots?$$

Matrix Multiplication

- ▶ In general, $A \times B \neq B \times A$
- ▶ $(A \times B)^T = B^T \times A^T$

Matrix Multiplication Exercise

$$X = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & -1 \end{bmatrix} \quad Y = \begin{bmatrix} 3 & 1 \\ 0 & -1 \\ -2 & 3 \end{bmatrix} \quad Z = \begin{bmatrix} 1 \\ 4 \\ 6 \end{bmatrix}$$

- ▶ $XY = \dots?$
- ▶ $YX = \dots?$
- ▶ $Z^T Y = \dots?$

Identity Matrix

- ▶ An identity matrix of size n is a $n \times n$ square matrix with ones on the main diagonal and zeros elsewhere.
- ▶ When an identity matrix is multiplied with another matrix A , the result is equal to A . It is analogous to multiplying a number by 1.

Example: Identity matrix of size 2 is given as,

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Identity Matrix

So, if we have matrix A ,

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\therefore I \times A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} = A$$

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Inverse of a Matrix

A inverse of a matrix, denoted as A^{-1} , is a matrix such that it satisfies the following condition:

$$A \times A^{-1} = A^{-1} \times A = I$$

where A is a $n \times n$ invertible matrix, and I is the identity matrix of size n .

Inverse of a Matrix

Think of it like a reciprocal of a number.

A number x and its reciprocal is given as $x^{-1} = \frac{1}{x}$ are multiplied together, the result is 1.

For example, $x = 2$

$$x \times x^{-1} = x \times \frac{1}{x} = 2 \times \frac{1}{2} = 1$$

Inverse of a Matrix

Inverse of a matrix is hard to compute by hand. But for a 2×2 size matrix, the formula is given as,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{a \cdot d - b \cdot c} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The matrix inverse does not always exist. Can you tell when that is the case for 2×2 matrices based on the formula given above?

Inverse of Matrix Application

When is matrix inverse useful? We can use it to solve systems of linear equations!

Consider the following equations:

$$x + 2y = 5$$

$$3x + 5y = 13$$

This can be written in a matrix form as,

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 13 \end{bmatrix}$$

Inverse of Matrix Application

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 13 \end{bmatrix} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 13 \end{bmatrix} \\ \therefore \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 13 \end{bmatrix} \end{aligned}$$

Now, we can easily solve the system of equations and get the solution for x and y

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Setting up Python

- ▶ Google Colab:
 - Interactive programming online
 - No installation
 - Free GPU for 12 hours
- ▶ Your task:
 - Register a Google account and set up [Google Colab](#)
 - Run `print('hello world!')`
 - Open Python Basics Demo from [Course Website](#)

Python Basics

- ▶ Program
 - We write operations to be executed on variables
- ▶ Variables
 - Referencing and interacting with items in the program
- ▶ If-Statements
 - Conditionally execute lines of code
- ▶ Functions
 - Reuse lines of code at any time

Python Basics

- ▶ Lists
 - Store an ordered collection of data
- ▶ Loops
 - Conditionally re-execute code
- ▶ Strings
 - Words and sentences are treated as lists of characters
- ▶ Classes (advanced)
 - Making your own data-type. Functions and variables made to be associated with it too.

Python Basics

- ▶ Write a function to find the second largest number in a list (Hint: use `sort()`)
- ▶ Define a class `Student`
- ▶ Use the `__init__()` function to assign the values of two attributes of the class: `name` and `grade`
- ▶ Define a function `study()` with an argument `time` in minutes. When calling this function, it should be printed “(the student’s name) has studied for (time) minutes”

NumPy Basics

A Python package is a collection of code people wrote for other users to run directly. Today, we learn how to use the package NumPy for linear algebra.

- ▶ Open NumPy Basics Demo from [Course Website](#)
- ▶ Your task: use NumPy functions to compute the exercises we did earlier this morning. Compare the results.